

Diffusion Models: Theory and Algorithms - NotebookLM

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I am taking a quiz on this material and was given this question: "In the derivation of the reverse conditional probability $p(x_{t-\Delta t} | x_t)$, a first-order Taylor expansion of $\log p_{t-\Delta t}(x_{t-\Delta t})$ around x_t is utilized. This approximation is crucial for relating the reverse process to the score function. Which of the following expressions correctly represents this Taylor expansion, and what is its primary consequence for the derivation?"

I chose the correct answer: "The expansion is $\log p_t(x_t) + \langle \nabla_x \log p_t(x_t), (x_{t-\Delta t} - x_t) \rangle + O(\Delta t)$. Its consequence is introducing the score function, which guides the reverse step toward regions of higher data density."

Help me understand more about this topic.

NotebookLM's quizzes have a nice feature where you can ask it to explain the correct answer, even if you got it right. In my case, I did get the first quiz question right because I knew the score function had to be involved somewhere, but I wanted to see the derivation again and check it with my attempted derivation. The explanation is quite thorough and largely uses information from the lecture notes without hallucination.

That is an excellent question that goes straight to the mathematical core of how Diffusion Models (specifically DDPM and related models) formulate their reverse process.

Here is a breakdown of the derivation and the importance of that first-order Taylor expansion, drawing directly from the provided source material:

1. The Purpose of the Reverse Conditional Probability

In diffusion models, the goal is to reverse the noisy forward process (the "Encoder") to generate clean samples (the "Decoder"). The reverse process involves stepping backward in time, from a noisy state x_t to a slightly less noisy state $x_{t-\Delta t}$.

The key distribution needed for this reverse step is the reverse conditional probability, $p(x_{t-\Delta t} | x_t)$. This distribution is necessary to transition accurately backward in time.

2. Derivation Using Bayes' Rule and Logarithms

The derivation starts by applying Bayes' rule to the reverse conditional probability:

$$p(x_{t-\Delta t} | x_t) = \frac{p(x_t | x_{t-\Delta t}) p_{t-\Delta t}(x_{t-\Delta t})}{p_t(x_t)}$$

Taking the logarithm yields:

$$\log p(x_{t-\Delta t} | x_t) = \log p(x_t | x_{t-\Delta t}) + \log p_{t-\Delta t}(x_{t-\Delta t}) - \log p_t(x_t)$$

We know that the forward step $p(x_t | x_{t-\Delta t})$ is typically a simple Gaussian distribution, which results in a quadratic loss term (like a squared L2 norm) in the log domain. However, calculating the exact marginal densities $p_{t-\Delta t}(x_{t-\Delta t})$ and $p_t(x_t)$ is intractable.

3. The Role of the Taylor Expansion

To make the term $\log p_{t-\Delta t}(x_{t-\Delta t}) - \log p_t(x_t)$ manageable, the derivation utilizes a Taylor expansion to relate the densities at adjacent time steps. Since Δt is assumed to be small, $\log p_{t-\Delta t}(x_{t-\Delta t})$ can be approximated around x_t :

The expansion you chose as correct is:

$$\log p_t(x_t) + \langle \nabla_x \log p_t(x_t), (x_{t-\Delta t} - x_t) \rangle + O(\Delta t)$$

This expression allows the original log reverse probability equation to be simplified by replacing the marginal density term, $\log p_{t-\Delta t}(x_{t-\Delta t})$, with an approximated version that uses the value of the density at time t ($\log p_t(x_t)$) and its gradient.

4. The Primary Consequence: Introducing the Score Function

The crucial consequence of this Taylor expansion is the introduction of the term $\nabla_x \log p_t(x_t)$. This term is mathematically defined as the **score function** of the data distribution at time t .

When the Taylor expansion is substituted back into the log reverse probability equation, and subsequent simplifications and approximations are made (including using the known Gaussian form of the forward process), the reverse conditional probability $p(x_{t-\Delta t} | x_t)$ is found to be proportional to a Gaussian distribution, $N(x_{t-\Delta t}; \mu_z, \sigma_q^2 \Delta t)$.

The mean μ of this reconstructed Gaussian distribution for the reverse step is directly determined by the score function:

$$\mu \propto x_t - \sigma_q^2 \Delta t \cdot \nabla_x \log p_t(x_t)$$

(This is derived by completing the square using the quadratic term from the forward process and the linear term from the Taylor expansion)

.) This means the reverse step is fundamentally directed by the score function, $\nabla_x \log p_t(x_t)$. **Interpretation:** The score function acts as a **directional vector field** that points toward regions of higher density of the true data distribution $p_t(x_t)$. **During the reverse (denoising) process**, the model uses a trained neural network f_θ to estimate this score function (or an equivalent noise term derived from it). This estimate effectively guides the sample x_t along the gradient toward the structure of the original data manifold, removing noise in a controlled manner.

In essence, the Taylor expansion allows the complex reverse step calculation to be approximated by a combination of the forward noise process (which is known) and the gradient of the log-likelihood. This substitution transforms the intractable reverse problem into a tractable estimation problem.

I am taking a quiz on this material and was given this question: "The document shows that after applying Bayes' rule and several approximations, the reverse conditional probability $p(x_{t-\Delta t} | x_t)$ can be approximated by a Gaussian distribution $N(x_{t-\Delta t}; \mu, \sigma_q^2 \Delta t)$. By completing the square in the log-probability expression, the mean μ is determined. What is the final derived expression for this mean, and what is its conceptual interpretation?"

I chose the correct answer: " $\mu \approx x_t + \sigma_q^2 \Delta t \cdot \nabla_x \log p_t(x_t)$. This represents a small step from the current noisy state x_t in the direction of the score function, effectively pushing the sample towards a region of higher data density."

Help me understand more about this topic.

While I also got this next question correct, I wanted to see a more in-depth explanation to verify my understanding. It was helpful to see the mathematical derivations support my intuitive understanding of the problem.

This topic delves into the core mechanism of how diffusion models reverse the noise process, effectively establishing the connection between the continuous mathematics of score-based models and the discrete steps taken during sampling.

Here is a detailed breakdown of the derivation and the conceptual meaning of the mean μ :

1. The Derived Expression for the Mean μ

The final expression derived for the mean of the approximate reverse conditional distribution, $p(x_{t-\Delta t}|x_t) \approx \mathcal{N}(x_{t-\Delta t}; \mu, \sigma_q^2 \Delta t)$, is indeed:

$$\mu \approx x_t + \sigma_q^2 \Delta t \nabla_x \log p_t(x_t)$$

This expression results from minimizing the log-probability of the reverse step, which, due to the combination of the forward process (a Gaussian) and the Taylor expansion, ends up having a quadratic form that can be written as the exponent of a Gaussian distribution.

2. The Process of Completing the Square

The derivation requires completing the square in the logarithmic probability expression, which combines two terms derived from Bayes' Rule and the approximations discussed previously:

1. **The Forward Step Term (Quadratic):** This comes from $\log p(x_t|x_{t-\Delta t})$, which is a Gaussian (Langevin dynamics or similar forward process):

$$-\frac{1}{2\sigma_q^2 \Delta t} \|x_{t-\Delta t} - x_t\|_2^2$$

2. **The Score Function Term (Linear):** This comes from the first-order Taylor expansion of $\log p_{t-\Delta t}(x_{t-\Delta t})$ around x_t :

$$\langle \nabla_x \log p_t(x_t), (x_{t-\Delta t} - x_t) \rangle$$

The goal is to reshape the combined terms into the canonical quadratic form of a Gaussian exponent, which is $-\frac{1}{2\sigma^2} \|x - \mu\|_2^2$.

The key manipulation involves grouping the terms involving $x_{t-\Delta t}$:

$$\log p(x_{t-\Delta t}|x_t) \propto -\frac{1}{2\sigma_q^2 \Delta t} (\|x_{t-\Delta t} - x_t\|_2^2 - 2\sigma_q^2 \Delta t \langle \nabla_x \log p_t(x_t), (x_{t-\Delta t} - x_t) \rangle)$$

By completing the square for the vector variable $x_{t-\Delta t}$ (using the identity $\|A - B\|^2 = \|A\|^2 - 2\langle A, B \rangle + \|B\|^2$), the expression transforms into:

$$\log p(x_{t-\Delta t}|x_t) \propto -\frac{1}{2\sigma_q^2 \Delta t} \|x_{t-\Delta t} - (x_t + \sigma_q^2 \Delta t \nabla_x \log p_t(x_t))\|_2^2 + C$$

Comparing this final form to the general definition of a Gaussian $\log \mathcal{N}(x; \mu, \Sigma)$, we clearly identify the mean μ as the term being subtracted from $x_{t-\Delta t}$ inside the squared norm:

$$\mu \approx x_t + \sigma_q^2 \Delta t \nabla_x \log p_t(x_t)$$

3. Conceptual Interpretation

The mathematical derivation here closely follows the lecture notes, but has some more steps to help me see all steps clearly. It then goes into a conceptual interpretation. I thought the dual explanation (both math and conceptual) was very helpful.

The conceptual interpretation is fundamental to understanding how diffusion models work during the generation (Decoder/reverse) phase.

A. The Role of the Current Sample (x_t): The step begins at the current noisy sample, x_t .

B. The Directional Force (Score Function): The critical component is the score function, $\nabla_x \log p_t(x_t)$. This term represents the **gradient of the log-probability density** of the data distribution $p_t(x_t)$ at the current time step t . Since the gradient points in the direction of the steepest ascent, the score function guides the sample toward regions where the underlying data density is higher.

C. The Drift Term ($\sigma_q^2 \Delta t \nabla_x \log p_t(x_t)$): This entire term determines the mean drift velocity for the reverse step.

- The factor Δt ensures that the drift is a small, instantaneous step backward in time.
- The factor σ_q^2 (variance) scales the movement, which is necessary to balance the inherent noisiness of the system.

In summary: The mean μ for the reverse step dictates that the new, slightly less noisy sample $x_{t-\Delta t}$ is obtained by taking the current noisy sample x_t and adding a correction step. This correction step is precisely engineered to follow the gradient of the data density function, which removes noise and pushes the sample toward the structure of the data manifold.

This process transforms the purely random motion (the forward diffusion process) into a guided trajectory that converges upon a clean sample, x_0 . The neural network trained for diffusion generation is essentially tasked with learning to estimate this score function (or an equivalent noise estimate related to it) at every time step t .

Analogy: Imagine standing blindfolded on a rugged mountain (the data manifold), trying to find the highest peak (the region of highest data density). If you randomly walk around, you might never find it. The mean μ for the reverse process tells you where to step: it takes your current position (x_t) and adds a push ($\sigma_q^2 \Delta t \nabla_x \log p_t(x_t)$) that is precisely determined by the local slope (the score function), ensuring that every small step you take is slightly uphill, thus guaranteeing you eventually find the peak.

NotebookLM also likes to give analogies to help understand how the score function travels slightly uphill at every step.

I am taking a quiz on this material and was given this question: "Claim 1 asserts that the reverse conditional $p(x_{t-\Delta t} | x_t = z)$ is approximately Gaussian with a variance of $\sigma_q^2 \Delta t$. What is the critical insight provided by the fact that this variance is fixed and identical to the variance of a single forward process step?" I chose this as the answer: "It indicates that the variance of the generated samples will always be equal to σ_q^2 , regardless of the number of steps taken." That answer was incorrect. The correct answer is "It means the amount of randomness in each reverse step is predetermined and does not depend on the specific location x_t in the data space, simplifying the learning task to only estimating the mean of the reverse conditional." Help me understand why my answer was incorrect.

This is a crucial conceptual point in understanding the implementation and efficiency of diffusion models, specifically models like DDPM (Denoising Diffusion Probabilistic Models). The reason your answer was incorrect relates to distinguishing between the variance of a **single step** and the variance of the **final output distribution**.

Here is a detailed explanation of why the correct answer is favored, based on the mechanics described in the sources:

Why Your Answer Was Incorrect I realize now why my answer was incorrect. NotebookLM focuses more on explaining why the actual answer is correct which is helpful.

Your chosen answer, "It indicates that the variance of the generated samples will always be equal to σ_q^2 , regardless of the number of steps taken," is incorrect because:

1. **Confusion of Step Variance vs. Terminal Variance:** The term $\sigma_q^2 \Delta t$ represents the variance added *during a single, infinitesimally small step* (Δt) of the forward or reverse process. The final generated sample x_0 must ultimately conform to the complex, typically non-Gaussian variance of the real data distribution $p(x_0)$.
2. **Cumulative Variance:** The final variance of the generated sample x_0 is the result of many steps, where the variance is accumulated or controlled across the entire reverse trajectory (from x_T down to x_0). While σ_q^2 often relates to the terminal noise variance $x_T \sim N(0, \sigma_q^2)$, it is not the fixed variance of the final clean sample x_0 .

Why the Correct Answer is the Critical Insight

The correct answer addresses the **computational simplification** that results from the fixed variance: **"It means the amount of randomness in each reverse step is predetermined and does not depend on the specific location x_t in the data space, simplifying the learning task to only estimating the mean of the reverse conditional."**

This is the critical insight for training the model:

1. **Decoupling the Drift (Mean) and the Diffusion (Variance):**The reverse conditional distribution is approximated as $p(x_{t-\Delta t} | x_t = z) \approx N(x_{t-\Delta t}; \mu_z, \sigma_q^2 \Delta t)$.
- The Mean (μ_z):** The mean μ_z (the "drift" or directional component) depends on the current sample location $z = x_t$ and the score function $(\nabla_x \log p_t(x_t))$, meaning it is **data-dependent** and must be learned by the neural network f_θ .
 - The Variance ($\sigma_q^2 \Delta t$):** The variance term $\sigma_q^2 \Delta t$ is a constant scaling factor for the noise $\eta \sim N(0, \sigma_q^2 \Delta t)$, which is used in every step of the reverse sampling process. It is **data-independent**. The explanation here isn't too short or too long. It uses both what it knows about diffusion to help explain the answer to me, and tells me where in the lecture notes this concept is illustrated.
2. **Simplification of the Learning Objective:**Because the noise schedule (the variance) is fixed, analytical, and known beforehand, the neural network f_θ does not need to learn the second-order statistical properties (the variance/covariance matrix) of the reverse process. Instead, the model is simplified and only needs to learn to predict the **mean** (the guidance or "denoising vector") at each step t . This makes the training objective much more stable and computationally feasible.

The sources illustrate this explicitly in the DDPM sampling pseudocode, where the reverse step $x_{t-\Delta t}$ is calculated using the learned function $f_\theta(x_t, t)$ (which predicts the mean) and then adding a fixed, pre-determined noise term η :

$$x_{t-\Delta t} \leftarrow f_\theta(x_t, t) + \eta, \quad \text{where } \eta \sim N(0, \sigma_q^2 \Delta t)$$

I am taking a quiz on this material and was given this question: "The pseudocode for DDPM sampling describes the reverse step as $x_{t-\Delta t} \leftarrow f_\theta(x_t, t) + \eta$, where $\eta \sim N(0, \sigma_q^2 \Delta t)$. In parallel, the DDIM sampling step is given as $x_{t-\Delta t} \leftarrow x_t + \lambda \alpha_t (f_\theta(x_t, t) - x_t)$. What fundamental difference in the generative process is captured by the contrasting update rules?"

I chose the correct answer: "DDPM is a stochastic process due to the addition of the random noise term η , generating slightly different samples each run, while DDIM is a deterministic process that produces the exact same sample given the same initial noise."

Help me understand more about this topic. I asked to explain further for this problem to have a more indepth explanation of the difference between DDIM and DDPM

The difference between the DDPM (Denoising Diffusion Probabilistic Model) and DDIM (Denoising Diffusion Implicit Model) update rules captures a fundamental split between **stochastic** and **deterministic** generative processes.

Here is a detailed breakdown of why the chosen answer is correct, based on the mechanisms shown in the sources:

1. DDPM: A Stochastic Process I wish this answer looked at the math a bit more, as it was mostly a conceptual explanation with some equations mentioned. However, the model is trying to answer using only the lecture notes which could explain that as the notes didn't have much on the math behind DDIM. It only had the pseudocode, which the answer referenced and explained mostly.

The DDPM update rule, $x_{t-\Delta t} \leftarrow f_\theta(x_t, t) + \eta$, explicitly includes a newly sampled random variable η in every reverse step.

- Noise Term:** Pseudocode 2 for DDPM sampling shows that in each loop iteration, a noise term η is sampled from a Gaussian distribution $N(0, \sigma_q^2 \Delta t)$.
- Stochasticity:** The addition of this freshly sampled noise η means that the trajectory taken from x_t to $x_{t-\Delta t}$ is *probabilistic*. Even if you start two sampling runs with the exact same initial noise $x_1 \sim N(0, \sigma_q^2)$, the sequence of generated samples $x_T, x_{T-1}, \dots, x_0$ will differ slightly because a new random η is drawn at every intermediate time step.
- Modeling:** DDPM directly models the reverse process $p(x_{t-\Delta t} | x_t)$ as a stochastic process governed by a Gaussian mean (the drift f_θ) and fixed, additive noise (the diffusion η).

2. DDIM: A Deterministic Process

The DDIM update rule, $x_{t-\Delta t} \leftarrow x_t + \lambda(f_\theta(x_t, t) - x_t)$, does not include a term for sampling and adding new random noise (like η) during the iterative reverse steps.

- Deterministic Trajectory:** Pseudocode 3 for DDIM sampling shows the update for $x_{t-\Delta t}$ relies solely on the current state x_t and the output of the trained model $f_\theta(x_t, t)$.
- Predictable Output:** If the process begins with a fixed initial sample x_1 (sampled once from $N(0, \sigma_q^2)$), the entire subsequent sequence of $x_{t-\Delta t}$ steps is determined purely by the fixed model f_θ and the known time schedule. Running the DDIM sampler multiple times with the same initial x_1 will always yield the identical final sample x_0 .

The fundamental difference lies in **how the model chooses its path backward in time**:

Feature	DDPM (Stochastic)	DDIM (Deterministic)
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Reverse Step	Follows the predicted drift plus random noise η .	Follows only the predicted drift/direction of the data manifold.
Sample Quality	Generates diverse samples from the full data distribution.	Produces a predictable sample trajectory given the starting noise.
Speed	Requires a high number of steps (T) to accurately model the reverse Markov chain.	Can generate high-quality samples using significantly fewer steps (e.g., non-Markovian skips) due to determinism.

The model is slightly misleading here by saying DDIM can generate high-quality samples, but it should have been more clear about DDPM generating more high-quality samples than DDIM as a result of having more steps.

I am taking a quiz on this material and was given this question: "According to Pseudocode 1, the training loss for the DDPM model is given by $\mathbb{E} \|f_{\theta}(x_{t+\Delta t}, t+\Delta t) - x_t\|_2^2$. What does this specific objective function imply about the architectural role of the neural network f_{θ} and its relationship to the derived mean μ of the reverse process?"

I chose the correct answer: "The network $f_{\theta}(x_{t+\Delta t}, t+\Delta t)$ is being optimized to directly predict x_t , which is the mean of the reverse conditional distribution $p(x_t|x_{t+\Delta t})$ when ignoring the stochastic term. Therefore, f_{θ} is a direct estimator of μ ."

Help me understand more about this topic.

This specific objective function is central to the design of Denoising Diffusion Probabilistic Models (DDPM) and fundamentally dictates what the neural network f_{θ} learns.

Here is a detailed explanation of the implication of the training loss $L \leftarrow \|f_{\theta}(x_{t+\Delta t}, t + \Delta t) - x_t\|_2^2$:

I wish this answer looked at the math a bit more, as it was mostly a conceptual explanation with some equations mentioned. However, the model is trying to answer using only the lecture notes which could explain that as the notes didn't have much on the math behind DDIM. It only had the pseudocode, which the answer referenced and explained mostly.

1. The Role of f_{θ} : A Denoising Autoencoder

The loss function $L \leftarrow \|f_{\theta}(x_{t+\Delta t}, t + \Delta t) - x_t\|_2^2$ trains the network f_{θ} to act as a **denoiser**.

- **Input to f_{θ} :** The input $x_{t+\Delta t}$ is a noisy sample created by adding noise to x_t (Line 4 of Pseudocode 1: $x_{t+\Delta t} \leftarrow x_t + N(0, \sigma_q^2 \Delta t)$).
- **Target:** The target is the original, less noisy state, x_t .
- **Objective:** The objective minimizes the squared L2 distance between the network's prediction, $f_{\theta}(x_{t+\Delta t}, t + \Delta t)$, and the true uncorrupted signal x_t . By minimizing this Mean Squared Error (MSE), the network is optimized to predict the mean (the expected value) of the distribution $p(x_t|x_{t+\Delta t})$.

2. f_{θ} as the Direct Estimator of the Mean (μ)

The critical implication is the relationship between the network's output and the derived mean (μ) of the reverse conditional probability:

- **Reverse Conditional Mean:** As established in our previous conversation, the reverse conditional distribution $p(x_{t-\Delta t}|x_t = z)$ is approximated by a Gaussian $N(x_{t-\Delta t}; \mu_z, \sigma_q^2 \Delta t)$. This mean μ provides the necessary "drift" or guidance to reverse the diffusion process.
- **DDPM Sampling Step:** Pseudocode 2 confirms that during generation, the new (denoised) state $x_{t-\Delta t}$ is calculated using the network's output as the mean of the step, plus noise (η): $x_{t-\Delta t} \leftarrow f_{\theta}(x_t, t) + \eta$.
- **Conclusion:** Because the DDPM sampling step uses $f_{\theta}(x_t, t)$ as the deterministic part of the reverse transition (the mean μ), the training objective, L , ensures that the network is learning to directly estimate this required mean μ (or a value that closely approximates it, x_t). The prediction of the uncorrupted sample x_t essentially is the mean of the reverse conditional distribution, μ , when the stochastic term (the noise η) is disregarded.

Conceptual Insight

The training process for DDPM is simplified because the model avoids the complex task of learning the score function $\nabla_x \log p_t(x_t)$ explicitly. Instead, **by training the network f_{θ} to simply denoise the image (predict x_t)**, the network implicitly learns all the information needed to calculate the mathematically required drift μ (which is closely related to the score function). This simplification makes the model highly efficient to train using standard MSE loss.

I am taking a quiz on this material and was given this question: "Figure 6 displays DDIM trajectories for a spiral dataset, noting that with more training data, the trajectories become 'more perpendicular to the underlying manifold.' What does this geometric property of the learned trajectories imply about the nature of the vector field the model has learned?"

I chose the correct answer: "It implies the model has learned the score function $\nabla_x \log p_t(x_t)$, whose vectors are gradients and are therefore orthogonal to the level sets (contours) of the probability density function, which locally define the manifold."

Help me understand more about this topic.

For this problem I wanted to know more about why increasing the number of samples made the diffusion look a lot better at a more in depth level rather than conceptual from lecture. Now, NotebookLM explains more about the score-based generative modeling which wasn't really mentioned in the notes.

This geometric property is the most direct visual evidence of the fundamental connection between diffusion models and the underlying concept of **score-based generative modeling**. The correct answer you chose captures this connection perfectly.

Here is a detailed explanation of why the perpendicularity of the trajectories implies the learning of the score function $\nabla_x \log p_t(x_t)$:

1. The Geometry of the Score Function (The Gradient)

The score function is mathematically defined as the gradient of the log probability density function at time t : $\nabla_x \log p_t(x_t)$.

- **Gradients and Level Sets:** In multivariate calculus, the gradient vector (∇) of a scalar field (like the probability density p_t) is always **perpendicular (orthogonal)** to the level sets or contours of that field [Information not directly present in sources, but essential to define "gradient" and "score function" as used in derivation].
- **Level Sets and the Manifold:** The data manifold—the underlying structure of the data (like the spiral in Figure 6)—can be thought of as existing along regions of extremely high probability density, p_0 . As noise is added during the forward process, the noisy data p_t spreads out, but it still concentrates around "ridges" or level sets that define the corrupted manifold.

• **Direction of the Score:** Since the score function points toward the direction of steepest density ascent (uphill), it is necessarily orthogonal to the curves of constant density.

2. The Role of the Score Function in the Reverse Process

As established in our previous discussion, the reverse process mean μ (the guiding vector for the reverse step) is determined by the score function:

$$\mu \approx x_t + \sigma_t^2 \Delta t \nabla_x \log p_t(x_t)$$

The reverse diffusion trajectory is essentially a path defined by following this drift term backward in time, which is driven by the score function. The DDPM training objective, $L \leftarrow \|f_\theta(x_{t+\Delta t}, t + \Delta t) - x_t\|_2^2$, optimizes the network f_θ to implicitly learn the score function by learning to denoise the data (i.e., predicting the mean μ , which is dependent on the score).

3. Interpreting the Perpendicular Trajectories

The comparison in the analysis to the data manifold which was mentioned in the lecture notes really helped tie in how the increased training data helped the model learn a vector field that points towards the data manifold. This was very insightful as I hadn't made this connection from just lecture alone.

Figure 6 visualizes the deterministic DDIM trajectories (which follow the mean drift of the learned vector field).

• **Observation:** As the model is trained with more data (N=10 to N=40), the reverse paths $x_t \rightarrow x_{t-\Delta t}$ become highly perpendicular to the spiral itself (the underlying manifold).

• **Implication:** This perpendicularity confirms that the model has successfully learned a vector field that points **straight toward the data manifold** (the highest density region), rather than moving along the manifold. The vector field being learned is the **score function**, which is characterized by this geometric property of being orthogonal to the level sets of the data density.

This geometric interpretation highlights the core principle of score-based modeling: samples are generated by initializing noise and then following the gradient of the data density until they collapse onto the data manifold in the denoised state (Phase 3: Clean Desired Samples).

Analogy: If the data manifold (the spiral) is like the bottom of a bowl, the level sets are the contours of the bowl. The score function is like the gravitational vector field that pulls everything straight down toward the bottom center. The reverse trajectories are the paths samples take as they are pulled out of the noise (the high part of the bowl) directly toward the manifold (the bottom of the bowl).

Overall, NotebookLM's explanations were quite insightful and helpful in further explaining confusing concepts that it quizzed me about. It mostly kept its explanations related to what was contained in the lecture notes, but expanded very well by going in-depth on concepts like the score function and even making connections back to the data manifold mentioned just at the start of the notes. This was very helpful in clearing my misconceptions both mathematically and conceptually in preparation for the final.